

EXERCISE – 8

Question

Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux Operating System:

- i) Detect the version of services running
- ii) Perform an aggressive scan
- iii) Detect the operating system of the target

Aim

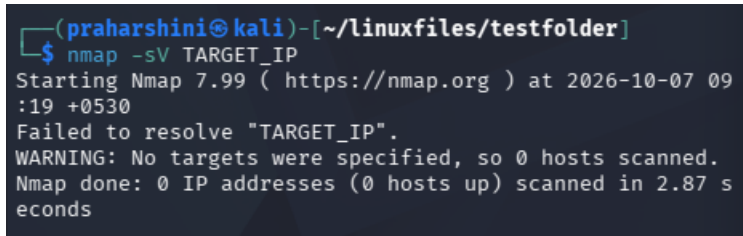
To perform Nmap service version detection, aggressive scanning, and operating system detection on a target system in Kali Linux.

Note: Use these scans only against a system you own or are explicitly authorized to test. Replace TARGET_IP with your authorized target's IP address.

1. Detect the Version of Services Running

Command:

```
nmap -sV TARGET_IP
```



```
(praharshini@kali)-[~/linuxfiles/testfolder]
$ nmap -sV TARGET_IP
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 09:19 +0530
Failed to resolve "TARGET_IP".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 2.87 seconds
```

Use

The -sV option is used to **detect the service and version information of open ports**.

2. Aggressive Scan Command

`nmap -A TARGET_IP`

```
(praharshini@kali)~[/linuxfiles/testfolder]
$ nmap -A 192.168.1.10
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 09:25 +0530
Nmap scan report for 192.168.1.10
Host is up (0.0018s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSSI S cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal)
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.50 ms 192.168.1.10

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.87 seconds
(praharshini@kali)~[/linuxfiles/testfolder]
```

Use

The `-A` option enables **Nmap's aggressive scan features** for more detailed target information.

3. Detect Operating System of the Target

Command

`nmap -O TARGET_IP`

```
(praharshini@kali)~[/linuxfiles/testfolder]
$ nmap -o TARGET_IP
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 09:23 +0530
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.04 seconds
```

Use

The `-O` option is used for **operating-system detection**.

Result

Thus, Nmap service version detection, aggressive scanning, and operating system detection commands were successfully executed in Kali Linux and the results were displayed